DRL Design Requirement Document

Team ID: 36 **Team Name:** *TECH-DECODERS*

School Name: *DBRA SOSE, SECTOR 11, ROHINI, DELHI 85*

Mentor Name: *Mr. Shubham Jain*

Team Members

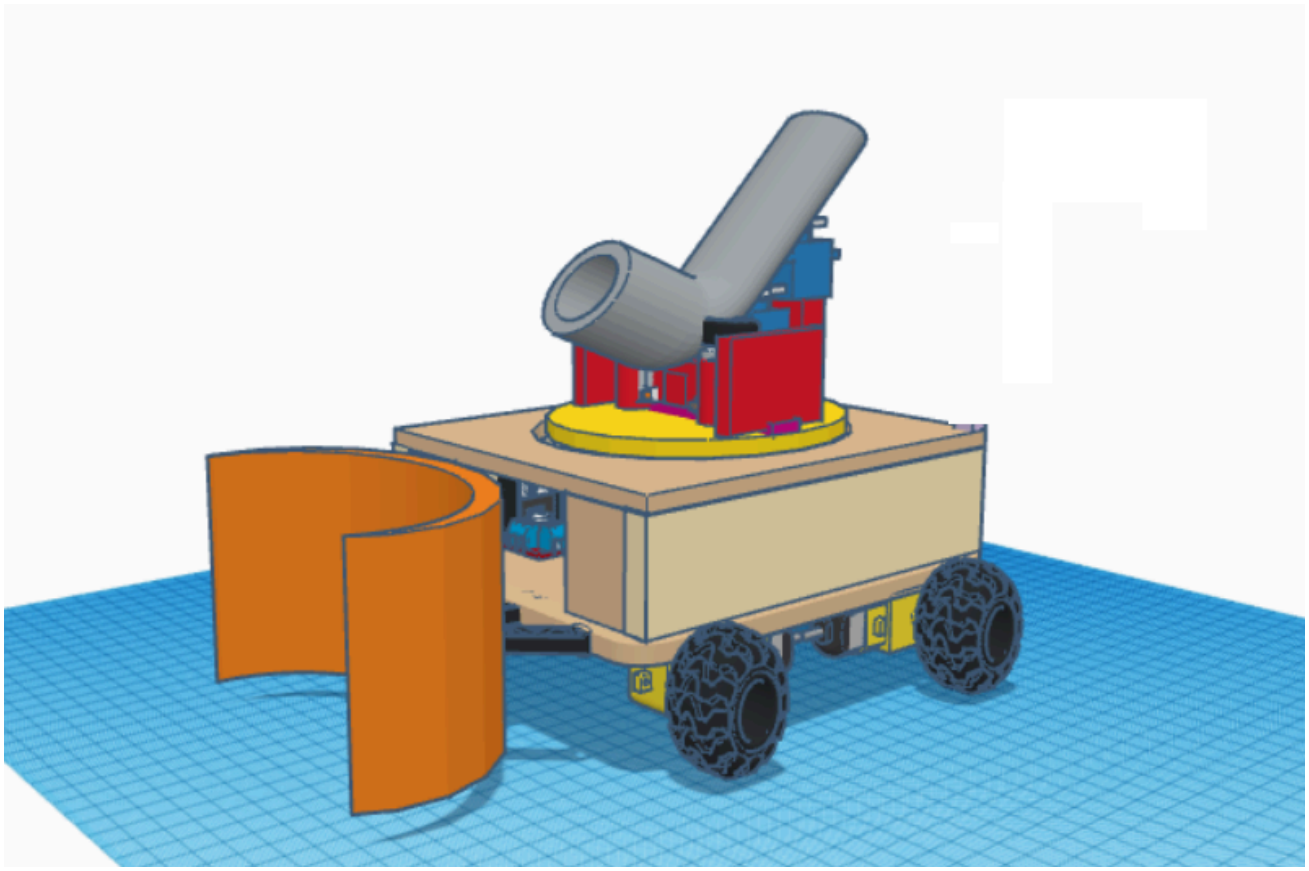
Main

1. Piyush
2. Gurmeet Singh
3. Ashmit Raj Nishad
4. Lavish
5. Chirag

Supporting

6. Gaurav Joshi
7. Vainessa Sharma
8. Ananta Rani
9. Aditya Maddheshiya

CAD/3D model of the robot



Robot different Parts	Dimensions
1. Grabber	135mm x 210mm x 215mm (LXBXH)
2. Chassis (main base)	400mm x 350mm Thickness 1.5cm
3. Shooting Pipe	70cm x 16cm (length X Diameter)
4. Left & Right Side	350mm x 1.5mm x 200mm (both sides)
5. Circular Disc	35cm Diameter 1.5cm Thickness
6. Overall Robot	485mm x 485mm x 800mm Weight 20kg(approx.)

Bill of Materials

S. No.	Name of the Component	Specifications	Justification	Quantity	Total Cost
1.	*Arduino Mega*	ATmega 2560	For Controlling Robot	1	***₹ 1700***
2.	*Arduino Nano*	ATmega328P.	For Remote Control	1	***₹ 2919***
3.	*NRF Transciever Module*	Nrf24L01+ *PA/ LNA*	For The Transmission of Data	1 Tx & 1 Rx	***₹ 369***
4.	*Motor Driver*	*L298N*	For Controlling All Motors	3	***₹ 873***
5.	*Bo Motors*	*single Shaft (500rpm)*	For Rotation of Mecanum Wheels	4	***₹ 220***
6.	*RS-555 DC Motor*	*12 Volt 5000 Rpm `*	For Shooting Mechanism	2	***₹ 440***
7.	*Servo Motor*	*MG995*	For Rotating The Circular Disc	1	***₹ 275***

8.	*Servo Motors*	*SG90*	For Rotation, Shooting Mechanism and Grabbing Mechanism	3	***₹ 320***
9.	*Joystick Module*	HW-504 Dual Axis XY	For Controlling By Remote	2	***₹ 200***
10.	*Potentiometer*	*Potentiometer 10K*	For Controlling the Voltage of 12v dc motors	1	***₹ 100***
11.	*Tactile Push button*	6mm x 6mm x 5mm	For Servo Motor Controlling of Shooting Mechanism and Grabbing Mechanism	4	***₹ 70***
12.	*Shooting Wheels*	*7cm diameter*	For Shooting Mechanism	2	***₹ 100***
13.	*Lipo Battery*	*12v 1100mah*	For Power Supply	3	***₹ 540***
14.	*BO Motor Mounting Clamper*		For Holding The Motor to the Base of the Chassis	4	***₹ 200***
15.	*Ply Board or Fibre Circular Disc*	*(Diameter 35cm)*	For Holding the Shooting Mechanism	1	***₹ 900***
16.	*Zero PCB*	*(9x15cm)*	For Making Remote Controller	3	***₹ 300***
17.	*Chassis(Aluminium or Steel)*	*500 X 500mm*	For Base and Covering	4	***₹ 940***
18.	*Jumper Wires All Type*			2(packs)	***₹ 200***
19.	*Mecanum Wheels with 7.5mm* *Couplers for Bo motors (pack of 4)*		For The Tyres of Robot Length – 3.1 inch or 80mm Breadth - 1.5 inch or 37mm	4	***₹ 800***

			Coupler – 7.5mm diameter		
20.	*PVC Pipe*	*(Diameter 16cm & Length 70cm)*	For Shooting Mechanism	1	***₹ 1000***
21.	*Zip Tie*	*10 inch*	For Mounting	20	***₹ 100***
22.	*Switch Button*	*12-24V*	For ON/OFF the robot	2	***₹ 120***
				Total	***₹ 12686***

S. No.	Name of Component	Specification	Justification	Quantity	Total Cost
1.	*Arduino Nano Cable*	*(USB Mini B)*	For Uploading Codes	1	***₹ 40***
2.	*Arduino Mega Cable*	USB Type (A to B)	For Uploading Codes	**1**	***₹ 40***
3.	*2m Red & 2m Black Wire*	22-25 AWG Wire	For Connections of DC Motor	**2 Packs Each**	***₹ 500***
4.	*DC Motor Mounting Clamper*	**Size:** 40 x 39 x 40mm (L x W x H) **Thickness:** 2mm **Mounting Holes Size:** 3.2mm **Base holes for Motor Shaft:** 17.2mm **Motor fitting holes size:** 4.2mm	For Holding The Motor to the Circular Disc	**2**	***₹ 40***

5.	*PVC Electrical Insulation Tape*		For Insulation	**1**	***₹ 20***
6.	*Rubber Sheet Pad*	500 x 500 mm	For Placing Between the Frame	**1**	***₹ 150***
7.	*Nut bolts with Washer*	Bolt Length : 8mm Diameter : 3mm	For Connections	**2 Packs**	***₹ 90***
8.	*Tapping Wood Screw*	Screw size : 50 x 8mm	For conecting the Frame	**2 Packs**	***₹90***
9.	*Aluminium Sheet*	*500 X 500mm*	For Shooting Frame	**4**	***₹ 940***
10.	*Lipo Charger*	*12V RC Pro Compact*	For Charging	**1**	***₹350***
				Total Cost	***₹ 2260***

Tools required for manufacturing the robot:

Sl. No	Tool Name (Hardware/Software)	Justification
1.	**Arduino Create cc (Software)**	**For coding the Arduino**
2.	**Aluminium Cutter**	**For cutting the aluminium sheets**
3.	**Multimeter**	**For measuring voltages etc.**
4.	**Screwdriver Kit**	**For tight and losing the screws**
5.	**Impact Drill Machine with Reversible Function and Drilling Accessories**	**For drilling**
6.	**Plier**	**For tight the nut bolts**

7.	**Soldering Machine**	**For solder on zero pcb**

Any deviation from Stage 1 submissions? If yes, elaborate with justifications:

Project Planning – Gantt Chart

Note: Individual columns represent weeks. Colour code the week blocks as per your plan. You may add/remove rows as req

	MONTH	APRIL					MAY					JUNE					JULY	
	WEEK NUMBER	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5	1	
ACTIVITY	START DATE	END DATE																
Design Finalization	--	17-04-2023																
Stage 1 results	17-04-2023	24-04-2020																
Collecting/Getting items	08-05-2023	21-05-2023																
Start working on robot	22-05-2023	31-05-2023																
Last checking of robot before stage -2 submission	01-06-2023	07-06-2023																
Stage 2 Submission	07-06-2023	09-06-2023																
Start using robot so we can control robot in the league easily	11-06-2023	18-06-2023																
Error checking and debugging.	19-06-2023	05-07-2023																
DRL Finals	06-07-2023	07-07-2023																